

1Voucher — Technical Docs

Operator reference for the platform’s value model, contracts, functions and the on-chain events that drive reporting. For a high-level product overview, see the **About** page.

Concepts Payments Gas fees Merchant Contracts Functions Events & Reporting

API

The three contracts

Contract	Purpose
Consumer	Smart-wallet factory + identity registry + append-only compliance log. Source of KYC level & country code that Vault/TreasuryToken read.
Vault	Unified balance ledger (ERC-4626 shares) for all currencies. Holds backing tokens; moves value; settles remittances and USD swaps; harvests yield.
TreasuryToken	Permissioned ERC-20 (e.g. TTZA) — the protocol’s closed-loop bank-cash token. Minted on fiat-in, burned on fiat-out. Compliance runs in its <code>_update</code> hook.

All three are UUPS upgradeable proxies; addresses are stable, implementations swap on upgrade (see the **Contracts** admin page for live versions).

Contract roles (on-chain)

A role is just a list of granted addresses. `initialize()` grants the **owner wallet** (`DEPLOYER_ADMIN_ADDRESS`) `DEFAULT_ADMIN_ROLE` + functional roles; the deploy script then `grantRole`s the operational roles to the **backend wallet** (`BACKEND_SIGNER_ADDRESS`) and `MINTER_ROLE` to the Vault. “Held by” = which of those was granted the role.

Contract	Role	Can do	Held by
TreasuryToken	<code>MINTER_ROLE</code>	Mint / burn (fiat deposit, settlement)	Owner, Vault, Backend
TreasuryToken	<code>PAUSER_ROLE</code>	Pause / unpause all transfers	Owner
TreasuryToken	<code>COMPLIANCE_ROLE</code>	Freeze/unfreeze, forced transfer (clawback/recovery), manage trusted whitelist	Owner, Backend
Vault	<code>ADMIN_EXECUTOR_ROLE</code>	<code>adminCredit/Debit</code> , <code>payRemittance</code>	Owner